



Does conditioning leave a structural memory?

Large-area evidence for the dislocation picture

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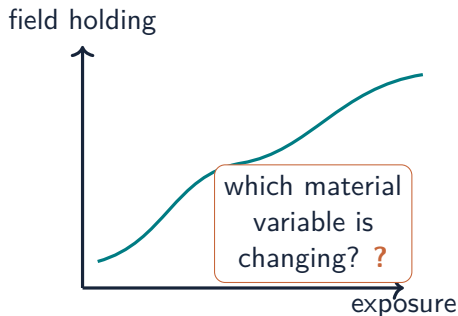
I. Popov (HUJI Nano) · W. L. Millar, V. M. Bjelland, W. Wuensch (CERN)



Conditioning works. What does it change?



- ▶ Conditioning is how every high-gradient device reaches its usable field.
- ▶ Breakdown rate has an extreme field dependence, $\sim E^{30}$.
- ▶ It tracks the number of **high-field pulses**, not the number of breakdowns.
- ▶ The field itself drives conditioning, not the arcs.



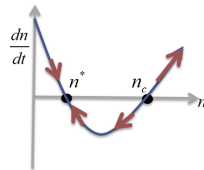
Something in the metal stores the improvement. We do not know what.

Source: Wuensch et al., Rev. Mod. Phys. 98, 025004 (2026); Degiovanni et al., PRAB (2016).

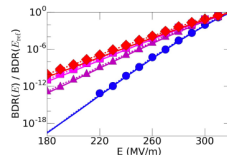
Our model: breakdown as collective dislocation dynamics



- ▶ Field holding rises FCC→BCC→HCP — the order of barriers to dislocation motion.
- ▶ The **MDDF** model: breakdown nucleates when the mobile-dislocation population, driven by Maxwell stress, runs away.
- ▶ One model gets the $\sim E^{30}$ rate, its temperature dependence, and the dark-current spikes.



mobile-dislocation runaway

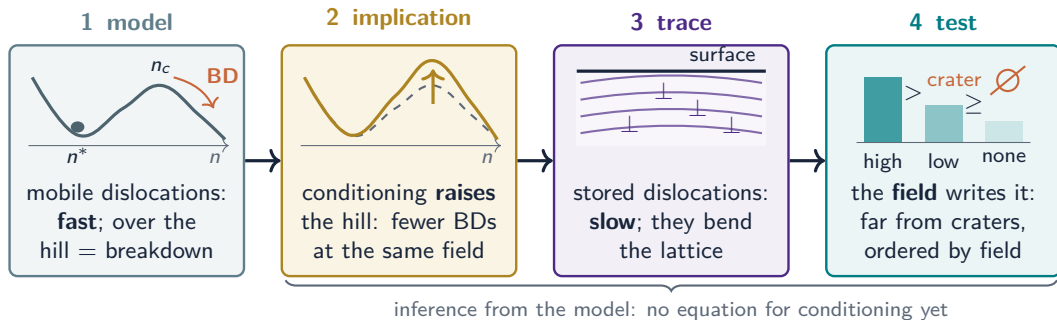


breakdown rate vs field

Central prediction: conditioning *evolves the near-surface dislocation structure*.

Source: Engelberg et al., PRL 120, 124801 (2018); PRAB 22, 083501 (2019); PRAB 23, 123501 (2020).

What the model says the microscope should see



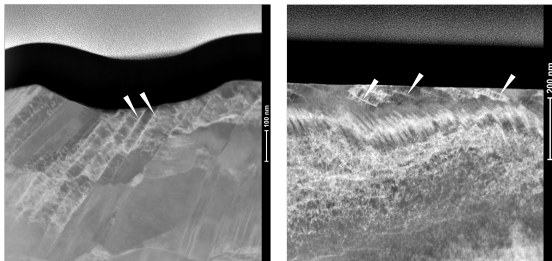
- ▶ In the model, a $\sim 10^9$ -pulse memory must sit in stored, EBSD-visible dislocations.
- ▶ The field writes it: look far from craters; expect the order to follow the local field.

predicts where, and in which order

does not predict size, depth, or sign

Source: Engelberg et al., PRL 120, 124801 (2018); PRAB 22, 083501 (2019); PRAB 23, 123501 (2020); Wuensch et al., RMP 98, 025004 (2026).

STEM already saw it, at single spots



Cross-sectional STEM of conditioned hard (as-machined) Cu: dislocation-poor zone in the top ~ 200 nm.

- ▶ **Direct** evidence of near-surface plasticity in conditioned regions.
- ▶ Fewer dislocation walls at 300 K and 30 K; denuded layer clearest at 30 K.
- ▶ Fits the clue that soft Cu conditions far more slowly than hard Cu: the starting population matters.
- ▶ FIB preparation plants its own dislocations; trust only large-scale structure.
- ▶ One cross-section cannot speak for the whole electrode.

Seen at one spot. The large-area comparison was still missing.

Source: Jacewicz et al., JAP 137, 193302 (2025); HUJI STEM (I. Popov), MeVArc 2025; Korsbäck et al., PRAB 23, 033102 (2020).

A tiny stress, repeated a billion times

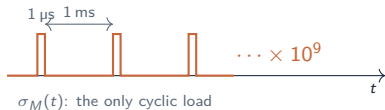


Maxwell stress

$$\sigma_M = \frac{\epsilon_0 E^2}{2}$$

at 80 MV/m: 0.028 MPa

$\sim 10^3 \times$ below the yield of annealed Cu



- ▶ Fatigue at 10^8 – 10^{10} cycles rearranges dislocations irreversibly, well below yield.
- ▶ Here: $\sim 10^9$ conditioning pulses.
- ▶ Pulsed DC means **no RF pulsed heating** and no bulk currents. Maxwell stress is the only cyclic load.
- ▶ MDDF is **threshold-free**: no minimum stress amplitude.

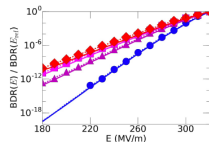
The model predicts a large-area signature. Last year we asked for it; this year we looked.

Source: Mughrabi, MMTB 40, 431 (2009); Stanzl-Tschegg et al., Int. J. Fatigue 29, 2050 (2007).

Last year we asked for more forensics. Here it is.

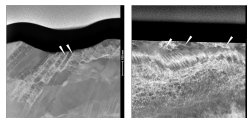


MeVArc 2025



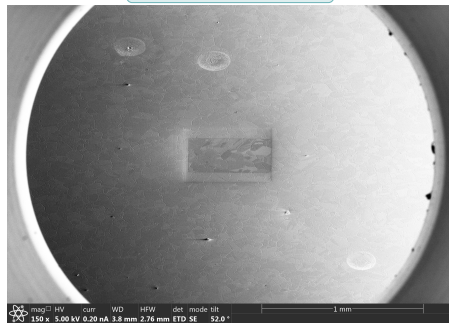
MDDF model

Closed with: other materials? more samples?



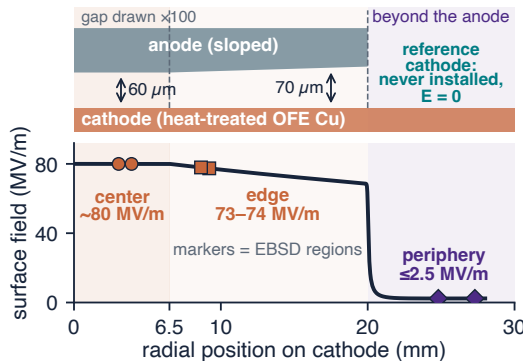
local STEM denuded zone

MeVArc 2026



Large-area EBSD across one cathode

One cathode turns field exposure into a controlled coordinate



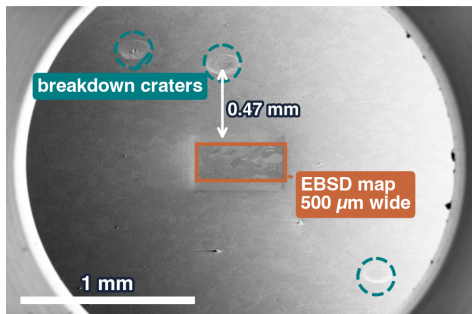
- ▶ Heat-treated OFE Cu, pulsed DC
- ▶ 1 μs pulses at 1 kHz, $\sim 10^9$ pulses
- ▶ Peak field ~ 80 MV/m at center
- ▶ Nine EBSD regions: center, edge, periphery, and external reference

● high field (center, edge) ● periphery (≤ 2.5 MV/m) ● unexposed reference

The field gradient gives us internal controls on a single specimen.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192 ($E(r)$: Fig. 3); geometry from Bjelland et al.

EBSD maps plastic activity far from breakdown craters



- ▶ Large area where TEM could only sample a point
- ▶ 500 μm -wide maps; 3 μm step (lateral only); information depth ~ 40 nm at 15 kV
- ▶ $\sim 24,000$ indexed points per ROI
- ▶ **LAM** = local average misorientation: lattice tilt between a point and its neighbors (KAM, LOS: cross-checks)
- ▶ Misorientation is a proxy for GND density

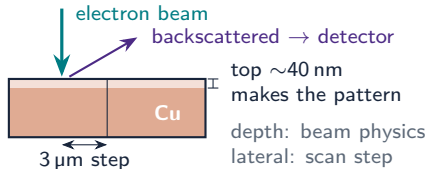
We measure the cumulative response to the field, not crater damage.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192; information depth ~ 40 nm at 15 kV: CASINO Monte Carlo, Drouin et al., Scanning 29, 92 (2007); measured 38–72 nm over 5–30 kV, Chen et al., Ultramicroscopy 111, 1488 (2011).

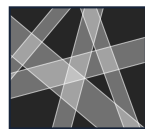
What one EBSD pixel measures



1 15 kV electron beam, point by point



2 Pattern → lattice orientation

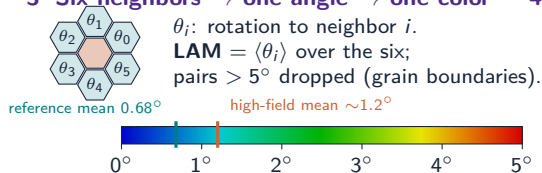


Kikuchi pattern

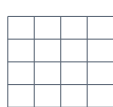


one lattice
orientation for
this point
~24,000 points
per region, unfiltered

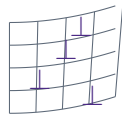
3 Six neighbors → one angle → one color



4 Why a rotation angle counts dislocations



flat: $\theta = 0$



curved: $\theta > 0$

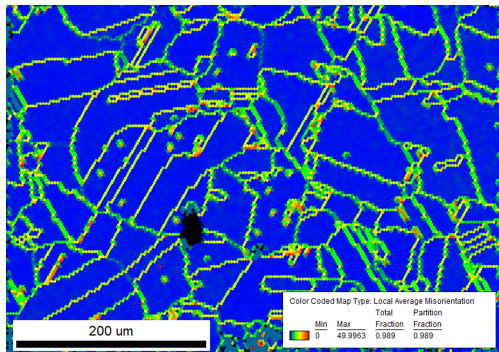
a dislocation proxy
over 500 μm , not a
dislocation image

Bending a crystal
by an angle over a
distance needs
**geometrically
necessary
dislocations**, so
**LAM is a proxy
for GND density.**

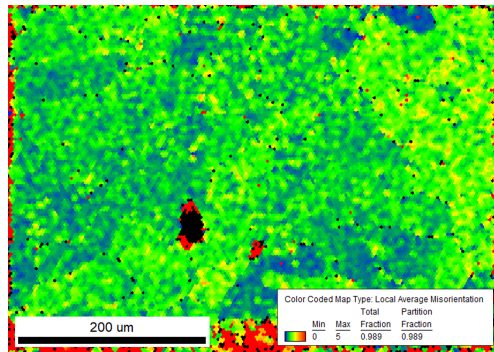
Each 3 μm point yields one angle, read from the top ~40 nm of copper.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192; information depth: Chen et al., Ultramicroscopy 111, 1488 (2011).

How to read a LAM map: wide-angle vs low-angle



0–50°: grain boundaries dominate

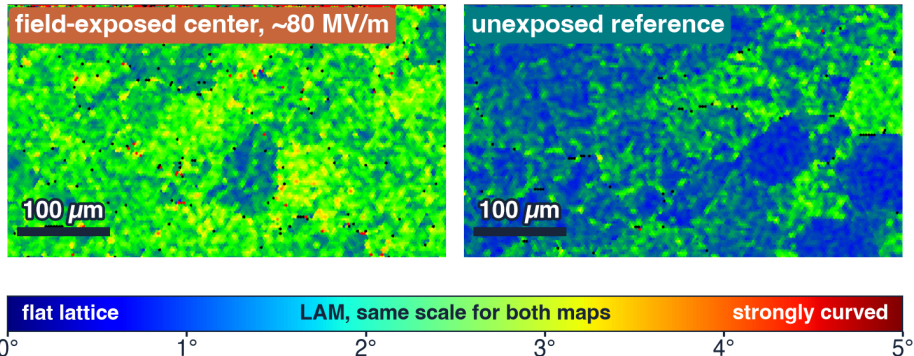


0–5° scale: intragrain curvature

Rescaling to 0–5° exposes the intragrain misorientation we care about.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192. Same area, two color scales.

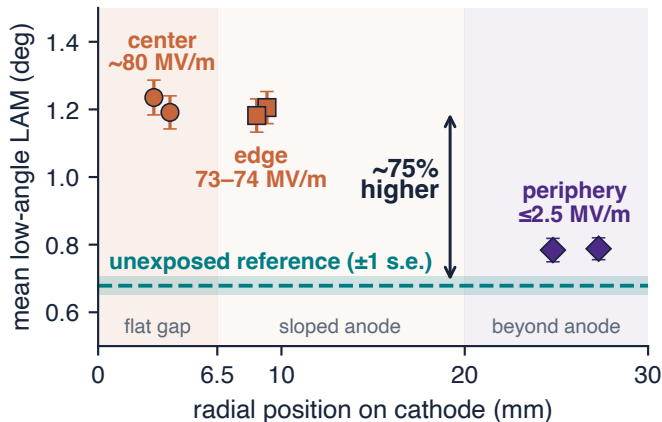
High-field copper contains more intragrain curvature



Same scale, same magnification: field-exposed grains carry more curvature.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192. Both maps cropped to the same $485 \times 279 \mu\text{m}$ field; LAM $0-5^\circ$.

The signal separates into three exposure tiers

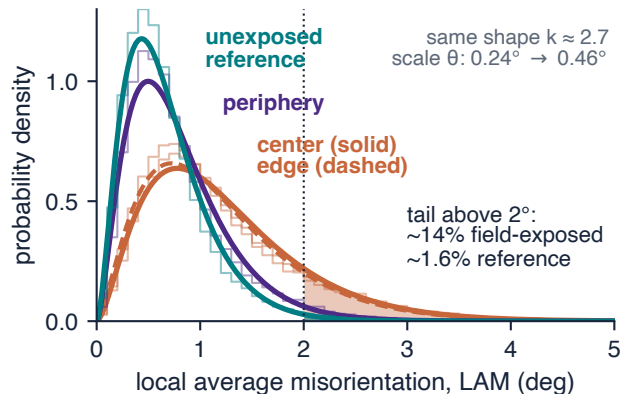


- ▶ Center and edge: $\sim 1.2^\circ$
- ▶ External reference: $\sim 0.68^\circ$
- ▶ High field is $\sim 75\%$ higher
- ▶ Grain size does not follow this hierarchy

The ordering follows field history, not sample position or grain size.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192. Error bars: grain-corrected standard errors; band: unexposed reference ± 1 s.e.

Field exposure builds a high-misorientation tail



LAM tail above 2°

~ 0.14 field-exposed
 ~ 0.016 reference

$\sim 8\times$ **increase**

Gamma shape $k \approx 2.7$ fixed; the scale roughly doubles.

The whole distribution stretches; a uniform offset cannot do that.

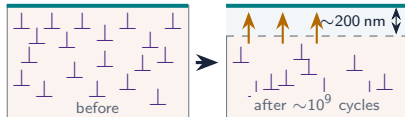
Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192. One ROI per tier (histograms faded, gamma fits solid, edge dashed); shaded: tail above 2° .

The large-area test supports the model's prediction



Hard, as-machined Cu

dense starting population (2025 STEM cathodes)



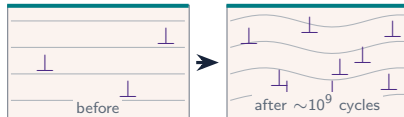
dislocations escape to the surface or annihilate $\Rightarrow$ **depleted layer** (STEM)

one load
 $\sigma_M = \frac{1}{2} \epsilon_0 E^2$
 $\sim 10^9$ cycles
 10^{-3} of yield

the manuscript's interpretation

Heat-treated OFE Cu

sparse starting population (this cathode)



same load generates and rearranges dislocations $\Rightarrow$ **lattice curvature** (EBSD, LAM $\sim 75\%$ higher)

- To our knowledge, the first **large-area** observation that conditioned regions carry a different near-surface dislocation structure, ordered as MDDF predicts.
- The three-tier ordering also tracks the conditioning variable E_S : a *candidate* structural basis for it.

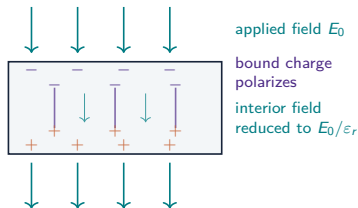
Before, dislocations were plausible. Now, a dislocation-sensitive state maps the conditioning history.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192 (discussion); Jacewicz et al., JAP 137, 193302 (2025).

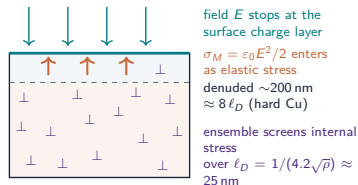
Elastic screening: dislocations respond like bound charge



Dielectric in an applied field



Copper under the Maxwell traction



applied field $\leftrightarrow$ Maxwell traction · bound charge $\leftrightarrow$ dislocations (dipole regime) · Debye length $\leftrightarrow \ell_D$ · screened interior field $\leftrightarrow$ screened internal stress of the depleted layer

2D single-slip theory at $\rho \sim 10^{14} \text{ m}^{-2}$; no theory yet treats cyclic loading with a free surface, so the $\sim 200 \text{ nm}$ thickness is observed, not predicted.

Charge screens the field at the surface. Dislocations screen the internal stress of a depleted layer, over $\sim 25 \text{ nm}$.

Source: Groma et al., PRL 96, 165503 (2006); Lemaître et al., PRE 104, 024904 (2021); Livne et al., PRE 107, 055004 (2023); PRAB, under revision; arXiv:2606.19192.

Copper establishes the effect, not universality



What is established

- ▶ Large-area structural difference
- ▶ Three independent metrics
- ▶ Internal and external controls
- ▶ Supports the MDDF prediction

What remains open

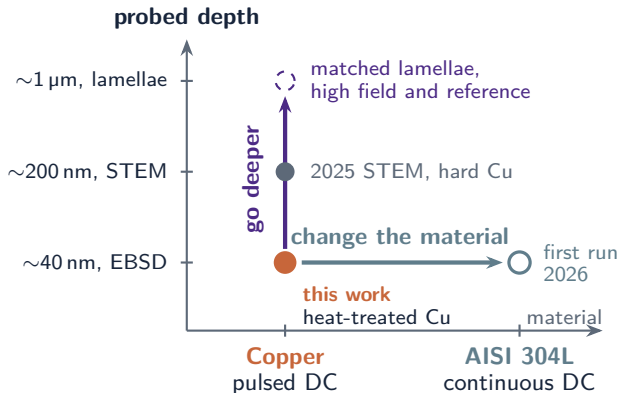
- ▶ One conditioned cathode, one experiment
- ▶ Full depth profile
- ▶ Grain-orientation dependence
- ▶ More conditioned cathodes
- ▶ Cross-material generality

Limits of this dataset: field level and breakdown density vary together by construction ($\sim 24, 12, 5$ breakdowns per cm^2 , center outward); LAM values are relative, compared only between regions mapped under the same step and protocol.

A stronger test changes both the material and the loading history.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192 (discussion, limitations).

Next: go deeper, and change the material



- ▶ Matched high-field and reference FIB lamellae; TEM/STEM + EBSD through the top $\sim 1\ \mu\text{m}$
- ▶ Artifact controls from the 2025 “red herrings”
- ▶ One depth profile from STEM depletion to EBSD curvature?
- ▶ Stainless steel under continuous DC, with Consorzio RFX — next slide

Depth continues the Uppsala STEM line; material tests the mechanism itself.

Source: Interim work plan (2026). Depths: EBSD $\sim 40\ \text{nm}$ at 15 kV (CASINO, Drouin 2007); hard-Cu STEM zone $\sim 200\ \text{nm}$ (HUJI, I. Popov 2025).

Steel: different material, different structural evolution



What differs from copper

- ▶ Lower stacking-fault energy: planar slip, different dislocation mobility
- ▶ Continuous DC: a steady traction of ~ 16 kPa, not $\sim 10^9$ pulses
- ▶ Metastable austenite is a built-in strain gauge: ε and α' martensite mark plastic strain above a few 0.1 %

First run with Consorzio RFX

- ▶ AISI 304L at the RFX HVPTF: continuous DC, 90 h and 73 h at 59–61 MV/m; zero breakdowns, stopped at emission switch-on
- ▶ 26 EBSD maps (HUJI, I. Popov): the exposed apex lies inside or below the unexposed range
- ▶ Nothing was mapped before exposure, so this design cannot establish a field effect



Consorzio RFX — De Lorenzi, Pilan, Spada

RFX HV program: N. Pilan, next talk (12:30).

Next campaign: one electrode mapped before and after being driven into breakdowns, with ε and α' in the phase list so that absent martensite becomes a strain bound.

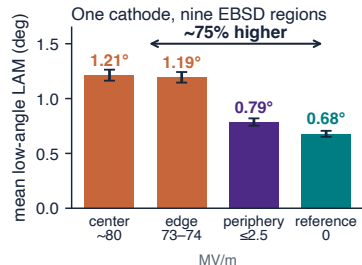
Copper supports the memory; steel asks whether it is universal.

Source: RFX project record, EBSD pass of 29 Sep 2026; ~ 16 kPa = $\varepsilon_0 E^2/2$ at 60 MV/m; strain gauge: Bayerlein et al., MSEA 114, L11 (1989).

Conditioning appears to leave a subsurface memory



1. Our model predicts conditioning evolves the near-surface dislocation structure.
2. 2025 STEM saw it locally; **large-area EBSD now supports it** ($\sim 75\%$ higher misorientation).
3. Depth and stainless steel test whether the memory survives a change of scale, alloy, and loading.



Last year: “other materials? more samples?” — here is the large-area answer.

Copper result: PRAB, under revision; arXiv:2606.19192. Full infographic in backup.

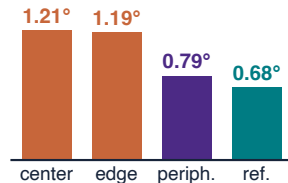
Conclusions, and an invitation



1. The model predicted a structural memory of conditioning.
2. STEM saw it locally; large-area EBSD now supports it.
3. Open: depth, replication, other materials — a stainless-steel test with Consorzio RFX is under way.

Talk to me: new materials, other electrode configurations, cross-facility comparisons, same-area before/after mapping — let us test this together.

mean LAM, one cathode



arxiv.org/pdf/2606.19192

Ashkenazy et al., PRAB, under revision;
arXiv:2606.19192

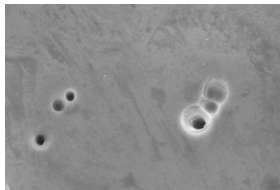
Source: Depth program: interim work plan (2026). Stainless-steel test: RFX project record (29 Sep 2026).

Backup: RFX first look (preliminary)

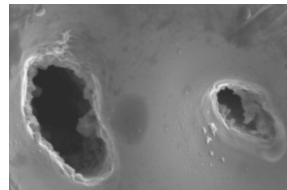


Preliminary — no field effect claimed

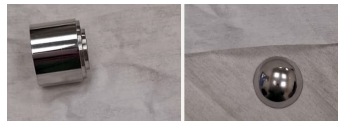
- ▶ AISI 304L at RFX HVPTF; continuous DC: 90 h (#3-25), 73 h (#4-25); peak 59–61 MV/m; zero breakdowns (stopped at emission switch-on)
- ▶ SEM: “exploded” near-surface inclusions (3–7 μm) on field-exposed samples; reference cavities round and clean
- ▶ EBSD: #1-25, #2-25, #3-25 index (97–99.5%); #4-25: two maps, one an outlier on confidence index, position unrecorded
- ▶ Apex-vs-side contrast reproduces on never-installed #2-25: forming geography



#1-25 reference: round, clean (HFW 16.6 μm)



#3-25 field-exposed: “exploded” (HFW 10 μm)



Sectioned electrode

Selected panels; the full SEM set is in the RFX SEM report.

Source: RFX project record through 29 Sep 2026; SEM report, Inna Popov (20 May 2026).

Backup: RFX sample and data status (29 Sep 2026)



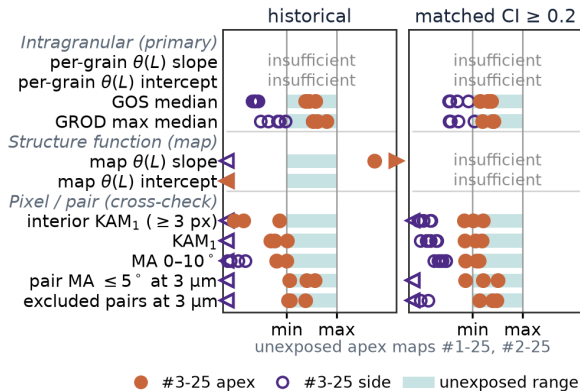
Sample	Role	Exposure	Status
#1-25	reference	none	SEM; EBSD apex + side, 97% indexed
#2-25	reference	none	EBSD apex (18 Aug, 3 maps) + side; LAM done
#3-25	field-exposed	90 h DC, peak 61 MV/m	SEM; EBSD apex + side, 99.5% indexed; LAM done
#4-25	field-exposed	73 h DC, peak 59 MV/m	SEM/EDS; EBSD two maps, one an outlier on confidence index

- ▶ Received since July: conditioning report and $E(\theta)$; alloy confirmed as 304L, mill certificate unrecoverable; electropolishing partly documented; all 26 catalog EBSD maps (36 scans), 28 Sep.
- ▶ XRD: δ -ferrite content differs between the reference and exposed pairs, so the clean comparison is apex vs side within one electrode.
- ▶ Wide-vs-narrow map class = grain size and indexing quality (also on never-installed #2-25), not field.
- ▶ Still open: thermal history of #3/#4 (vacuum firing?); matrix Cr/Ni EDS on all four under identical conditions; SEM θ -survey as the first exposure marker.

Backup: RFX apex vs unexposed range (29 Sep 2026)



Null so far — not a field effect, not proof of saturation

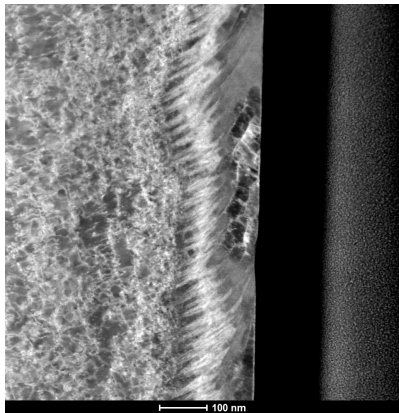


Wide map class, 2 of 4 panels (full board: RFX record). Only point above range: map $\theta(L)$ slope ($n = 2$ baseline).

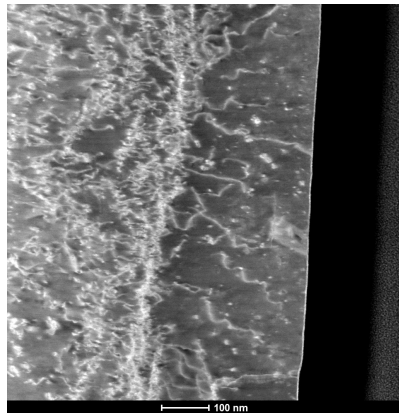
Source: RFX project record, orientation pass of 29 Sep 2026 (all 26 catalog maps); EBSD I. Popov, HUJI.

- ▶ Unit = one EBSD map; #3-25 apex vs same-class unexposed apex ($n = 2-3$ vs $3-4$).
- ▶ Inside or below the unexposed range (GOS, in-grain $\theta(L)$ slope, interior KAM, histogram indicators), also at matched confidence index.
- ▶ Where it differs it is *cleaner*, not rougher ($p \geq 0.10$).
- ▶ Power: 80% at 1 SD needs ≈ 17 maps per group; the data exclude only a uniform apex shift $\gtrsim 2.5$ map-SD. Runs stopped at emission switch-on, zero breakdowns.

Backup: field-exposed vs reference STEM (2025)



Field-exposed, 300 K: fewer walls



Reference, 300 K: walls to surface

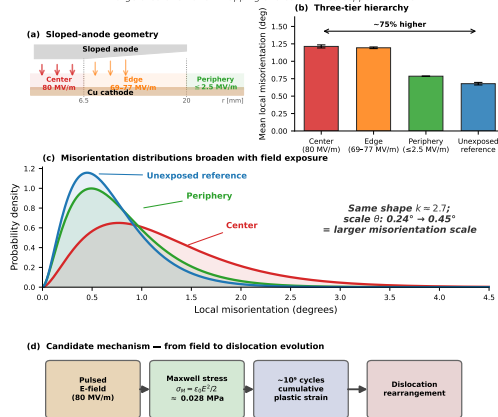
Source: HUJI STEM (I. Popov), MeVArc 2025; Jacewicz et al., J. Appl. Phys. 137, 193302 (2025), Figs. 8–10.

Backup: copper result in one figure



Field-Induced Subsurface Plastic Activity

Large-area orientation mapping of a conditioned copper cathode



First large-area observation of field-induced dislocation structure → candidate physical basis of the conditioning state E_s